



1. Audio Stem & Metadata Ingestion Pipeline

Step 1: Creator Uploads Audio Stems

Artist initiates upload on *tunemavens.com*. Frontend requests presigned S3 URL from FastAPI backend.

Step 2: Binary Stems Persisted to S3 / Cloudflare R2

High-definition audio stems (WAV/AIFF 24-bit) streamed directly to Object Storage.

Step 3: Metadata Indexing & ISRC Generation

FastAPI indexes title, splits, genre & ISRC code into MongoDB `catalogs` collection.

Step 4: Sync Clearance & Multi-Site Fanout

Catalog marked `is_cleared_for_sync: true` and synced to SyncMavens & TuneStream player.

2. Sync Licensing Match & Escrow Split Waterfall

Step 1: Music Supervisor Submits Brief

Supervisor posts project brief (budget, license type, genre) on *syncmavens.com*.

Step 2: AI Match Simulation & Pitch Submission

AI Match Engine scores catalog stems against brief. Creator approves pitch submission.

Step 3: E-Signature Execution & Stripe Escrow Deposit

Contract generated in `contracts`. Supervisor deposits license fee into Stripe Escrow.

Step 4: Automated Waterfall Payout Disbursed

Option 1: 30% Master / 25% Sync / 52.5% Writer; Option 2: 15% Agency Fee dispatches automatically.

3. Fan Lossless Streaming & Direct Tipping Engine

Step 1: Fan Listens on TuneStream

User streams lossless audio on *tunestream.co* web or native Capacitor app.

Step 2: Stream Event Logged in MongoDB

Playback event written to `activity_events` collection for real-time social graph recommendations.

Step 3: Fan Triggers One-Click Micro-Tip

Fan tips artist \$5-\$50 via Stripe payment widget directly from player interface.

Step 4: Immediate Creator Wallet Credit

Stripe Connect transfers funds directly to artist's linked bank account with 0% platform lockup.

4. Social AI Studio & EPK Marketing Pipeline

Step 1: Creator Prompt Input

Creator inputs visual concept or lyrics in Mother Platform *intermaven.io*.

Step 2: AI Multi-Format Generation

Gemini Nano generates cover art; Sora 2 renders 9:16 short vertical promotional video clips.

Step 3: Mother-EPK Auto-Hydration

Assets, catalog tracks & press stats compiled into interactive EPK grid link.

Step 4: CRM Marketing Campaign Dispatch

Resend API emails EPK press blast & push notifications to opted-in fans and industry leads.

Ecosystem Interaction & Administrative Control Matrix

User Role	Primary Target Frontend	Key Actions & Triggers	Administrative Control Panel	Admin Governance Capabilities	Target Database Collections
Creator / Artist	tunemavens.com intermaven.io	- Upload audio stems & metadata - Build Mother-EPK & Social AI assets - Configure split payment ledgers	Mother-CMS Admin DSP Catalog Admin	- Allocate monthly AI creation credits - Verify creator PRO registration status - Audit ISRC codes & catalog ownership	users catalogs publishing_deals
Music Supervisor	syncmavens.com	- Submit sync briefs & budget parameters - Review AI pitch match scores - Execute digital license contracts	Stripe Escrow Admin AI Governance Admin	- Approve supervisor enterprise access - Verify escrow deposit fund releases - Tune AI brief match algorithm weights	briefs pitches contracts
Fan & Listener	tunestream.co	- Stream 24-bit lossless audio tracks - Send direct fan tipping payments - Bookmark EPKs & follow creators	Mother-CMS Admin Stripe Escrow Admin	- Moderate public comment feeds - Monitor micro-tipping transaction velocity - Flag suspicious streaming behavior	activity_events messages users
Record Label / DJ Pool	djs.tunemavens.com corporate.tunemavens.com	- Manage roster catalog submissions - Download DJ pool edits & extended mixes - Execute bulk distribution deals	DSP Catalog Admin Stripe Escrow Admin	- Approve label account privileges - Audit multi-artist payout ledgers - Track DSP delivery (Spotify/Apple)	distribution_deals catalogs contracts
Platform Operator	All 4 Portals	- Overlook system operational health - Feature top artists on Wall of Fame - Audit compliance & copyright claims	All Admin Panels	- Global override & account freeze - Direct database query & split re-alignment - System campaign broadcast dispatch	All Collections

Security & SSO Token Architecture:
Single Sign-On is handled by an OIDC / PKCE PyJWT token engine running at auth.py. Session cookies are wildcard-scoped to .tunemavens.com, allowing seamless cross-app auth across all 4 sites without re-authenticating.

Financial Auditability & Escrow Protection:
All licensing payouts (Option 1: 30/25/52.5 and Option 2: 15% agency fee) and direct fan tips are backed by Stripe Connect double-entry ledgers. Admin approval is required for payouts exceeding \$10,000 to prevent fraud.